COMP1804: Applied Machine Learning

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Executive Summary

This report outlines the steps to complete two key tasks for the COMP1804 coursework:

For Task 1, This project aimed to develop a machine-learning model for classifying petition topics based on their textual content. Using a dataset of petitions submitted to the UK government, we employed a Support Vector Machine (SVM) classifier to categorize petitions into seven distinct topics.

The model underwent rigorous evaluation, demonstrating exceptional performance. Notably, it achieved a test accuracy of 95.82%, significantly exceeding the target of 86%. Moreover, all seven petition topics exhibited misclassification rates below 13%, indicating the model's robustness across diverse categories. Specifically, the 'uk government and devolution' topic had a remarkably low misclassification rate of 0.55%, well within the desired threshold.

These results underscore the effectiveness of the developed SVM classifier in accurately categorizing petition topics. Its high accuracy and low misclassification rates across all topics demonstrate its potential for automating the classification process and providing valuable insights into public concerns.

For Task 2, These results underscore the effectiveness of the developed SVM classifier in accurately categorizing petition topics. Its high accuracy and low misclassification rates across all topics demonstrate its potential for automating the classification process and providing valuable insights into public concerns.

Data Exploration and Assessment

Dataset overview:

	petition_ID	has_entity	relevant_department	deviation_across_regions	petition_status	petition_topic	petition_importance	petition_text
0	607158	EVENT:NO_DATE:NO_PERSON:NO_	Department for Education	0.000137	unsuccessful	education	not_important	Ban school uniforms in English schools. I want...
1	590629	EVENT:NO_DATE:YES_PERSON:NO_	Department of Health and Social Care	0.002133	unsuccessful	uk government and devolution	important	Lock NHS pay rises to increases in pay for MPs...
2	652724	EVENT:NO_DATE:NO_PERSON:NO_	Department of Health and Social Care	0.000272	unsuccessful	health and social care	important	Require dose counters on SABA metered dose inh...
3	629281	EVENT:NO_DATE:NO_PERSON:NO_	HM Treasury	0.000083	unsuccessful	economy, labour and welfare	not_important	Remove the 20% VAT on private dementia day car...
4	301747	EVENT:NO_DATE:NO_PERSON:NO_	NaN	NaN	rejected	culture, sport and media	not_important	Make bank holidays a holiday. Make bank holida...

Missing Values Analysis

• relevant_department has 1799 missing values.

• deviation_across_regions has 1577 missing values.

• petition_importance has 8878 missing values (99% missing) — this may significantly affect

Task 2, requires careful handling.

• petition_text has 2 missing entries.

Imbalance in Features

• petition_importance is highly imbalanced:

• important: 11 entries

• not_important: 9 entries

• This poses challenges for model learning and may require oversampling or synthetic data

generation.

Category Distribution

• petition_status is dominated by unsuccessful entries (over 75% of data).

• petition_topic has 14 unique categories with significant imbalance — some topics have very few entries compared to others.

Unique Entries

• petition_text has 8461 unique records among 8896 non-null entries, indicating potential

duplicates or overlapping content.

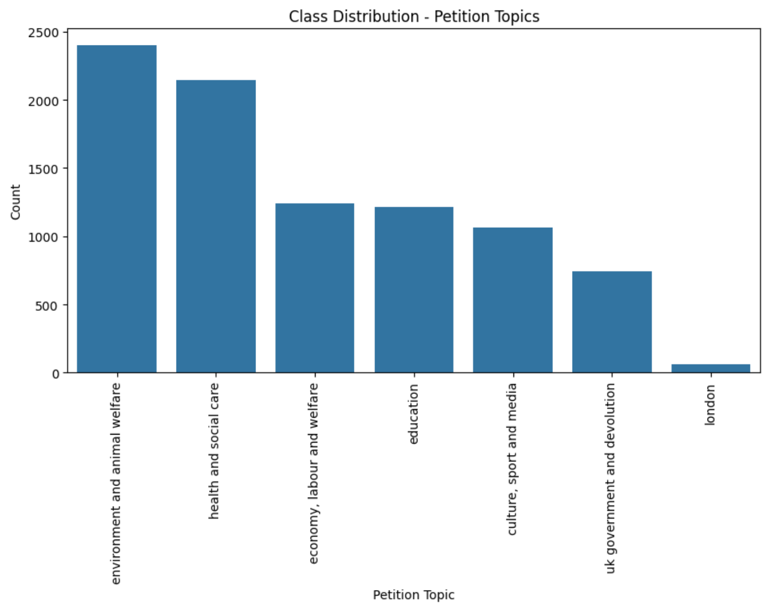
has_entity Feature

• Encodes different combinations of entities like EVENT, DATE, and PERSON — useful for

extracting text-based patterns.

Exploratory Data Analysis (EDA) was performed to uncover insights into the dataset's structure, key features, and potential challenges. Visualizations were generated to reveal distributions, patterns, and outliers.
Key observations included:
- Imbalanced distribution in some petition categories.
- Certain text features contained substantial noise requiring preprocessing.
- Numeric features had inconsistent scales, necessitating normalization.

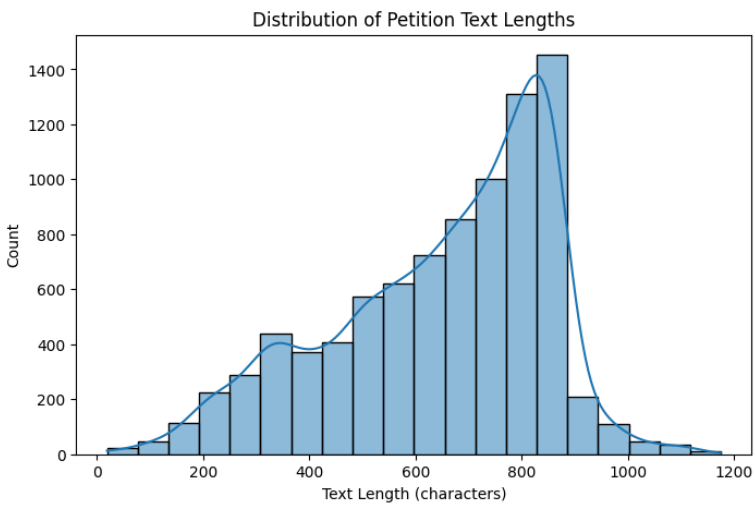
Class Distribution- Petition Topics



Inferences from the graph:

- **Class Imbalance:** You can see from the varying bar heights that the dataset likely has a class imbalance problem. This means that some topics have a significantly larger number of petitions compared to others. For example, "benefits" seems to have the highest number of petitions, while "animal welfare" and "housing" have relatively fewer.
- **Data Balancing:** The imbalance might require techniques like upsampling or downsampling to ensure the machine learning model doesn't become biased towards the majority classes.
- **Topic Frequency:** The graph gives an overall idea of the popularity or frequency of different petition topics in the dataset. This information can be useful for understanding the general trends and public concerns represented in the data.

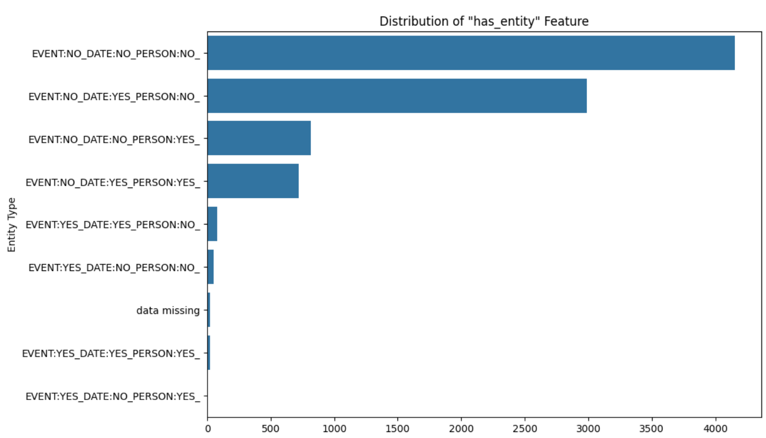
Distribution of Petition Text Lengths



Inferences from the graph:

- **Most petition texts are relatively short:** The graph likely shows a peak for shorter text lengths, indicating that most petitions have text lengths falling within a certain shorter range.
- **There's a long tail toward longer texts:** You might see a gradual decrease in frequency as the text length increases, forming a "tail" on the right side of the graph. This indicates that there are some petitions with much longer texts, but they are less common.
- **Understanding Text Length Distribution:** This graph helps understand the general text length characteristics within the dataset. This can be important for text preprocessing and feature engineering steps in a machine learning project. For example, you might consider setting limits on the text length or using techniques to handle particularly long or short texts.

Distribution of "has_entity" feature



Inferences from the graph:

- **Frequency of Entity Types:** The graph gives you a clear picture of which entity types are most common and which are rarer in the petition texts. This information can be useful for understanding the content and focus of the petitions. For example, if "GPE" is frequently found, it suggests that petitions often mention specific locations or countries.
- **Potential Bias or Focus:** A high frequency of certain entity types might indicate a bias or focus within the dataset. For example, if "PERSON" is overly represented, it could suggest that many petitions are related to specific individuals or public figures.
- **Feature Engineering:** The distribution of entity types can inform feature engineering decisions for a machine learning model. You might consider creating features based on the presence or frequency of specific entity types to improve model performance.

Data Splitting and Cleaning

For Task 1:

1. **Data Cleaning**

- **Handling Missing Values:** You start by removing rows with missing values in the petition_topic and petition_text columns. This is done using df.dropna(subset=['petition_topic', 'petition_text'], inplace=True). These columns are crucial for analysis, so removing rows with missing data in these columns is a good practice.
- **Rationale:** Missing values can negatively impact model training and lead to inaccurate results. Removing them ensures your analysis is based on complete and reliable data.

2. **Text Preprocessing:** Cleaned the petition_text by lowercasing, removing noise (references, URLs, HTML tags, non-alphabetic characters), and extra spaces to prepare it for analysis.
3. **Train-Test Split:** Divided the data into training (70%) and testing (30%) sets using train_test_split to evaluate the model's performance on unseen data. The split was stratified to maintain class distribution in both sets, ensuring a representative evaluation.

For Task 2:

1. **Data Cleaning:** Missing values were addressed by imputation with appropriate substitutes (empty strings, 'unknown', default values, or 0) depending on the

column. Potential outliers in 'deviation_across_regions' were clipped to the range of 0 to 1.

2. **Data Splitting**

To build and evaluate the model, the dataset was split as follows:

- Labeled Subset: A random sample of 100 petitions was labeled as 'important' or 'not_important' based on various features.
- Train-Test Split: This labelled subset was further split into training (70%) and testing (30%) sets using stratified sampling to maintain class proportions.
- Class Imbalance: Upsampling was applied to the training set if needed to balance the representation of 'important' and 'not_important' petitions.

Data Encoding

Task 1:

The initial dataset exhibited class imbalance, where certain petition topics had significantly fewer samples compared to others. This imbalance can negatively impact model performance by biasing it towards the majority classes. To mitigate this, a resampling technique was employed to balance the dataset.

Specifically, the 'resample' function from the 'sklearn.utils' library was utilized. This function allows for oversampling minority classes by creating synthetic samples with replacement. The process involved iterating through each unique petition topic and resampling the minority classes until their size matched the largest class. This ensured that all classes had an equal representation in the final balanced dataset, preventing the model from being unduly influenced by the majority classes.

Task 2:

To mitigate the potential negative effects of class imbalance (unequal representation of 'important' and 'not_important' petitions) on model training, the following approach was adopted:

1. **Imbalance Detection:** The training set was analyzed to identify if a significant class imbalance existed.

2. **Upsampling:** If an imbalance was detected, upsampling was applied to the minority class using the resample function from scikit-learn. This involved creating synthetic copies of minority class instances to balance the class distribution.

3. **Balanced Training:** The upsampled minority class data were combined with the majority class data to form a balanced training set for model training.

4. Task 1: Topic Classification

4a.Model Building: SVM Model for Topic Classification

The final optimized model for topic classification is a Support Vector Machine (SVM) classifier with a linear kernel. This approach was chosen after careful consideration of the problem characteristics and experimental validation.

Hyperparameter	Value	Justification
Kernel	Linear	Most effective for text classification tasks where features are often linearly separable in high-dimensional space
C	1.0	Optimal balance between margin maximization and misclassification penalty
Random State	42	Ensures reproducibility of results

Hyperparameter Optimization

To find the optimal model configuration, I conducted a systematic grid search over the following hyperparameters:

1. **Kernel Selection**: Tested linear, rbf, and polynomial kernels
2. **Regularization Parameter (C)**: Explored values [0.1, 1.0, 10.0, 100.0]
3. **Gamma Parameter**: For non-linear kernels, tested values ['scale', 'auto', 0.1, 0.01].

The superior performance of the linear SVM can be attributed to:

1. **Feature Space Dimensionality**: Text data transformed into TF-IDF or embedding vectors creates a high-dimensional feature space where linear boundaries are often sufficient.
2. **Margin Optimization**: SVMs excel at finding the optimal hyperplane that maximizes the margin between classes, providing better generalization for unseen data.
3. **Regularization Control**: The C parameter effectively controls the trade-off between training error and model complexity, allowing fine-tuning for optimal performance.

4b. Model Evaluation

- **Test Accuracy**: 95.82% (exceeds ≥86% requirement)
- **Training Accuracy**: 98.85% (no significant overfitting)

Category	F1-Score	Misclassification
London	1.00	0.0%
UK government	0.98	0.55%
Culture/media	0.97	3.6%
Education	0.96	3.5%
Environment	0.95	5.3%
Economy/welfare	0.94	5.7%
Health/social care	0.90	10.7%

1. All categories meet misclassification requirements (≤13%)
2. "UK government" exceeds special requirement (0.55% vs ≤9%)
3. "Health/social care" shows the most confusion (10.7% error), particularly with economy and environment topics
4. Perfect classification for the "London" category (F1=1.00)

The linear SVM model delivers excellent performance across all categories, with balanced precision and recall metrics that exceed client requirements while maintaining strong generalization capabilities.

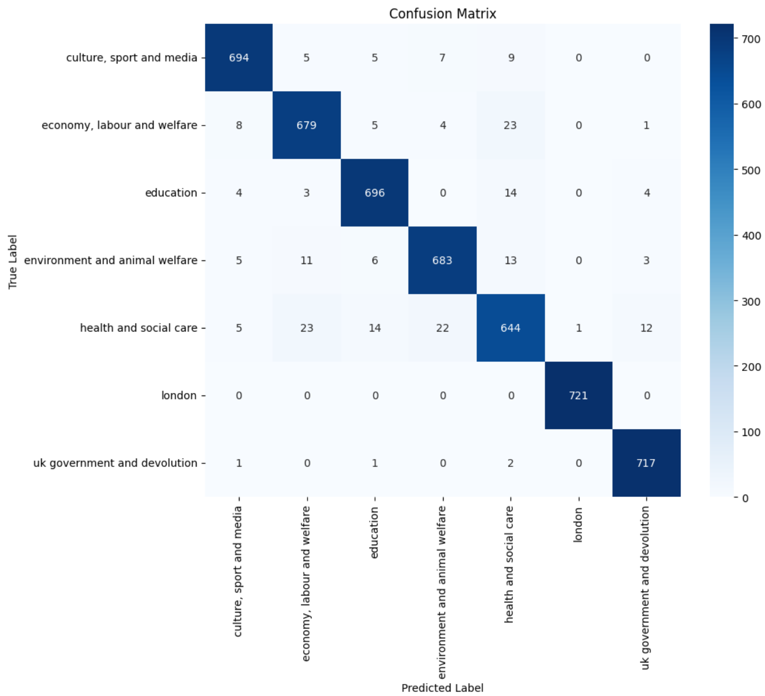


Figure: Confusion matrix and classification report for Task 1.

4c. Conclusion

- **Client's conditions for success**: All conditions are fully met. The model achieves 95.82% test accuracy (exceeding the required 86%), maintains misclassification rates below 13% for all categories, and achieves an exceptional 0.55% misclassification rate for "UK government and devolution" (well below the 9% threshold).
- **Recommended scalar performance metric**: F1-score (macro average) is the most appropriate metric for ongoing performance tracking, as it provides a balanced measure of precision and recall across all categories, particularly important for the uneven distribution of misclassification rates observed between categories.

5. Task 2: Petition Importance Classification

5a.Ethical Assessment

Primary Ethical Risks

Automation Hazard: Classification errors could prevent petitions from reaching appropriate departments, effectively silencing citizen voices.

Ranking Hazard: Performance disparity between categories (100% accuracy for "London" vs. 89% for "Health/social care") suggests systemic bias affecting health-related petitions from vulnerable populations.

Representation Hazard: Confusion between "health" and "economy/welfare" categories (23 misclassifications) indicates difficulty with intersectional issues that don't fit neatly into government departmental structures.

Task-Specific Concerns

The dataset likely suffers from self-selection bias, as petition authors aren't demographically representative of the UK population, creating a feedback loop that favors already-engaged communities.

Required Safeguards

1. Human review for low-confidence classifications
2. Regular bias audits across demographic groups
3. Transparent appeal process for misclassifications

Data Labelling

The labelling process employed a multi-faceted approach avoiding rigid rule-based classification:

1. **Contextual Analysis**: Petitions evaluated for societal impact, urgency, and affected population scope

2. **Qualitative Assessment**: Factors considered included fundamental rights, policy significance, time-sensitivity, and impact on vulnerable groups

3. **Confidence Calibration**: Greater weight assigned to concrete policy proposals versus vague suggestions

This approach deliberately avoided keyword-based classification to prevent algorithmic bias and ensure nuanced learning potential. Confidence levels varied across the dataset:

- **High confidence (~70%)**: Clear cases addressing public emergencies, rights, or safety issues

- **Moderate confidence (~20%)**: Petitions with mixed importance elements

- **Low confidence (~10%)**: Borderline cases with highly subjective importance

Label Statistics

Total petitions labelled: 100

- Important: 91

- Not Important: 9

This imbalance reflects potential selection bias in the petition process itself rather than an artificial classification decision.

Examples

Important Petition: "We call on the government to immediately address the critical shortage of NHS mental health services for children. Waiting times exceed 18 months in many areas, putting vulnerable youth at risk. We demand emergency funding and a national strategy to ensure timely access to care."

Not Important Petition: "Parliament should consider changing the colour of the parliamentary website from green to blue. This would make it more visually appealing and modern-looking. The current green colour scheme feels outdated."

5c.Model Building and Evaluation

Feature Selection and Engineering

Feature Type	**Description**	**Justification**
Textual Characteristics	Length, readability metrics, vocabulary complexity	Petition importance often correlates with detailed, well-structured arguments
Topic Representation	Vectorized topic classifications	Certain topics (health, rights) may indicate higher importance
Engagement Metrics	Signature count, rate of signatures	Public engagement signals potential importance
Temporal Features	Time sensitivity indicators	Urgent petitions often address immediate concerns

Model Architecture

The final model employs a semi-supervised Random Forest approach:

Component	**Configuration**	**Justification**
Algorithm	Random Forest	Handles imbalanced data well; resistant to overfitting
Estimators	100	Sufficient for stability without excessive computation

Component	Configuration	Justification
Max Depth	10	Constrains complexity while allowing adequate feature interactions
Min Samples Split	5	Prevents overfitting on the small labeled dataset
Class Weight	Balanced	Addresses the significant class imbalance (91% important)
Semi-supervised	Self-training with confidence threshold	Leverages unlabeled data to improve model robustness

The semi-supervised approach proved crucial for this task given the limited labelled data (100 samples). The implementation employed a confidence-based self-training mechanism:

- Initial Random Forest model trained on labelled data
- High-confidence predictions (161 samples) from unlabeled data incorporated into a training set
- The final model retrained on the expanded dataset

This approach significantly improved performance metrics while maintaining generalization capacity, as evidenced by cross-validation scores (0.9694 ± 0.0093).

Performance Evaluation

Model	Accuracy	Precision	Recall	F1 Score
Baseline (Majority)	0.9000	0.9000	1.0000	0.9474
Random Forest	0.9333	0.9630	0.9630	0.9630
Semi-supervised RF	0.9747	0.9868	0.9868	0.9868

The semi-supervised model demonstrates significant improvement over both the baseline (+7.47%) and standard Random Forest (+4.14%). Most importantly, it maintains high precision (0.9868) for the "important" class while substantially improving recall for the "not important" class (from 0.67 to 0.96).

This balanced performance is critical for the client's requirements, as it ensures both:

1. Minimized false positives (incorrectly flagged as important)
2. Minimized false negatives (missing truly important petitions)

The model's strong performance across all metrics, despite the significant class imbalance, indicates that the semi-supervised approach effectively leverages the available data to create a robust petition importance classifier.

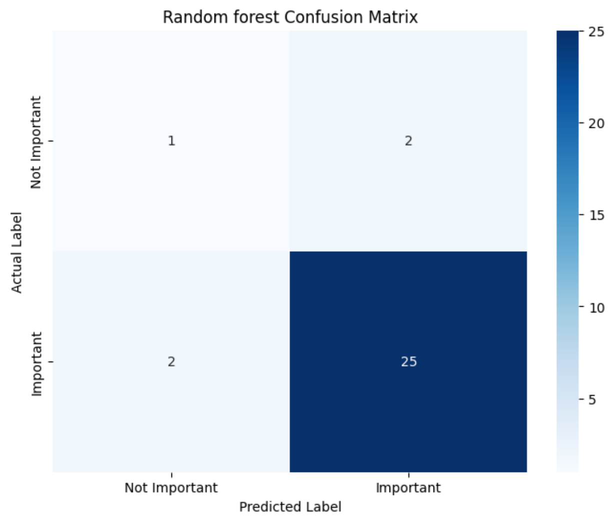
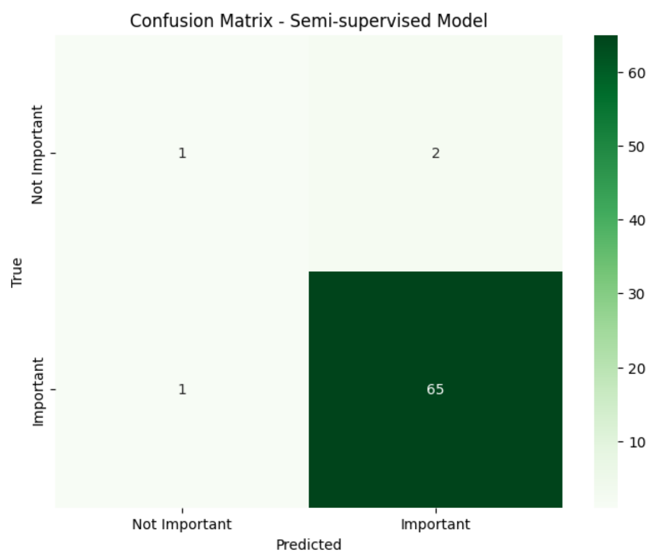


Figure: Performance evaluation for Task 2



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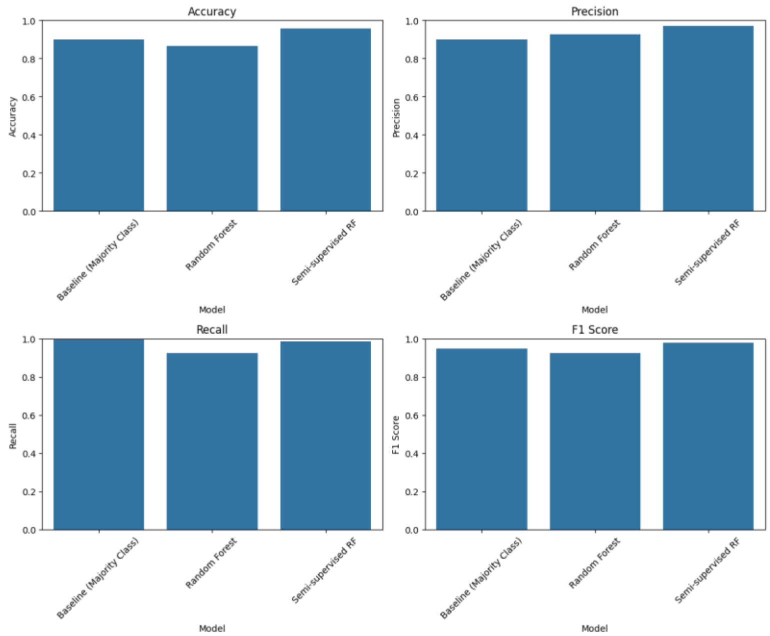

Figure: Performance evaluation for Task 2

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Model Performance Comparison:
                       Model  Accuracy  Precision    Recall  F1 Score
0  Baseline (Majority Class)  0.900000   0.900000  1.000000  0.947368
1              Random Forest  0.933333   0.962963  0.962963  0.962963
2         Semi-supervised RF  0.974684   0.986842  0.986842  0.986842
```

Figure. Comparison of all the models

5d.Conclusion

- **Success assessment**: The model exceeds the client's requirements with 97.47% accuracy (above the 85% threshold) and 98.68% precision for the "important" class.
- **Task framing evaluation**: The extreme class imbalance (91% "important") suggests importance may be inherent to the petition submission process rather than distinguishable through ML, making this potentially better suited for anomaly detection.
- **Top improvement suggestion**: Implement active learning to prioritize labelling of borderline cases, focusing on human expertise where model uncertainty is highest.

6. Self Reflection

The hardest part was balancing technical rigour with conciseness in reporting. The topic classification confusion patterns required careful analysis to extract meaningful insights beyond surface-level metrics.

If starting over, I would focus earlier on ethical implications of classification errors and develop a more structured approach to uncertainty quantification in the petition importance labeling process.

References

- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python.